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Mutual Adaptation and Influence: Survey of Latent Dynamics Models in Human-Robot Interaction

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Keywords

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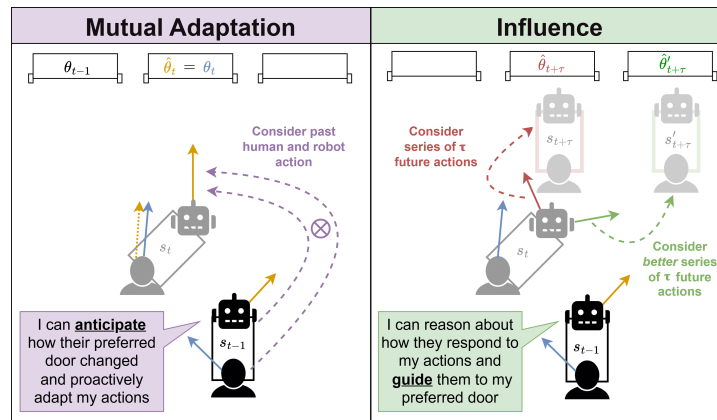
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Abstract:

Recent advances in robotics have enabled more dynamic and interactive roles when working with humans. This is partially afforded by the use of latent states, which allow the robot to predict human actions in a compact and tractable way. This paradigm has allowed robots not only to respond and adapt to human behavior but also to anticipate and plan proactively through the use of latent dynamics models. This survey explores the use of such models in the field of human-robot interaction (HRI). Current works revealed three classes of latent dynamics models: Bayesian, Markovian, and encoded. To connect these works, we synthesize a unified framework consisting of a prediction and control step. This framework shows how the traditional one-way adaptation extends to mutually adaptive behaviors by using latent dynamics in the prediction step. Similarly, we show how influence is an extension of mutual adaptation by using latent dynamics in the control step. We then review state-of-the-art approaches to mutual adaptation and influence for the three latent dynamics classes and discuss emergent properties as they relate to application, update frequency, and specific considerations for mutual adaptation or influence. Using this review, we highlight gaps in the current literature and propose future directions.



ToC Figure: Robots are becoming better teammates by learning hidden patterns in how people act. This survey covers how state of the art approaches leverage these patterns to help robots not only react, but also anticipate and guide cooperation. Here, we synthesize a unifying framework for these approaches, classify and review existing works, and highlight key trends and open challenges in this field.

1 Introduction

With continued technological advances, the problem of *collaboration* between humans and robots in complex tasks has risen to the forefront. In collaborative interactions, humans and robots work together simultaneously on shared tasks, where team performance can be improved by modifying the robot's behavior to better coordinate with human actions [1]. To that end, it is essential for the robot not only to model the environment but also to model its human partner. Providing a robot with a theory of mind (ToM)—the ability to reason about human beliefs, desires, and goals—can significantly improve human-robot coordination [2]. The standard approach to planning in this setting is to modify robot behavior in response to some *latent state* that compactly represents the human's policy. By estimating the previous latent state from the previous interaction, the robot can better predict the future behavior of the human and plan supplemental actions [3]. We refer to this as *one-way adaptation*, where the robot creates a new, one-shot estimate at every timestep.

However, humans are often non-stationary and adapt to robot actions, which is a critical consideration for successful HRI [4]. The change in a human's latent state over time depends on past interaction with the robot, as well as on their previous latent state. We refer to the change of latent state over time as *latent dynamics*. A robot can leverage latent dynamics to better anticipate the latent state in the current interaction by reasoning about how past robot actions caused the previous latent state to evolve into the current one. Consequently, the robot can then use the improved prediction of the current latent state to plan actions that are better coordinated. We refer to this as *mutual adaptation*, which anticipates the current, unseen latent state—whereas one-way adaptation only estimates the previous latent state at best.

Not only can latent dynamics be used to anticipate the human's current latent state, it can also be used iteratively during planning of future actions. Thus, applying a model of latent dynamics to robotic control extends mutual adaptation to *influence*—a concept linked to improved performance in HRI [5]. Here, the robot incorporates latent dynamics into its control policy to anticipate how its actions will shape the human's future latent state. By doing so, the robot can guide the human to a latent state that results in a more desirable outcome. Although influence operates on the human's latent state, the objective may or may not include eliciting a particular latent state known in advance. Consequently, influence emerges naturally from planning with access to a latent dynamics model.

To further elaborate, consider the scenario in Figure 1, where a robot and a human attempt to move a table through one of three doors. Knowing which door the human prefers (latent state) can help the robot predict the force the human will apply to move the table (action), so that the robot can plan its action accordingly. Accurate prediction of this latent state is therefore essential for smooth and effective interaction. In the case of *one-way adaptation*, the robot can

observe the human's action in the previously observed state s_{t-1} to *estimate* which door the human preferred, denoted as θ_{t-1} . This estimate can then be used to modulate robot behavior in the current state s_t .

However, the human will likely observe the interaction forces that the robot previously applied in s_{t-1} , similarly to how the robot did in one-way adaptation. Consequently, the human may update their preferred door to θ_t based on this observation in order to adapt to the robot's behavior, which would make θ_{t-1} an outdated estimate. Thus, the robot acknowledging the human's update and adapting its behavior accordingly is referred to as *mutual adaptation*. This affords the robot the ability to better *anticipate* the human's preferred door in the upcoming interaction, compared to one-way adaptation, thereby improving the robot's action planning.

Taking this one step further, we can exemplify *influence*, where the robot takes a more active role in planning future team behavior by repeatedly reasoning about how the human would adapt their preferred door in response to the robot's future interaction forces. Assume that the robot has a vested interest in which door the table is carried through. By using its model of how the human updates their preference in response to the robot's actions, the robot can reason about what sequence of actions would lead the human to prefer door $\theta'_{t+\tau}$ instead of door $\theta_{t+\tau}$ after τ timesteps. The robot could then take the corresponding sequence of actions to *guide* the human to prefer the door that better suits the robot's needs (i.e., $\theta'_{t+\tau}$).

1.1 Objective

Due to the use of latent dynamics being somewhat novel and the diversity of domains in HRI (e.g., human factors [7], AI [8], and controls [9]), researchers have implemented varying methods in several applications (e.g., autonomous vehicles [10] and contact robots [11]) without a strong interconnecting theory. Consequently, there is a general lack of formalism for the distinction between mutual adaptation and influence, thereby causing these terms to often be used interchangeably.

The primary objective of this paper is to synthesize a unifying framework for mutually adaptive and influential models from the existing literature on non-stationary humans with latent dynamics models in HRI. This framework provides a clear system of organization in an attempt to reduce confusion caused by the conflation of these distinct approaches. Secondly, we bring attention to state-of-the-art research in this field while applying the proposed framework as an organizational tool. Thus, readers have access to a breadth of approaches spanning the full scope of the framework such that they can easily access information that suits their needs. Lastly, this review identifies future directions for this field in a way that is grounded and motivated through the reviewed works.

The remainder of the paper is organized as follows: Section 2 provides the background information required to main-

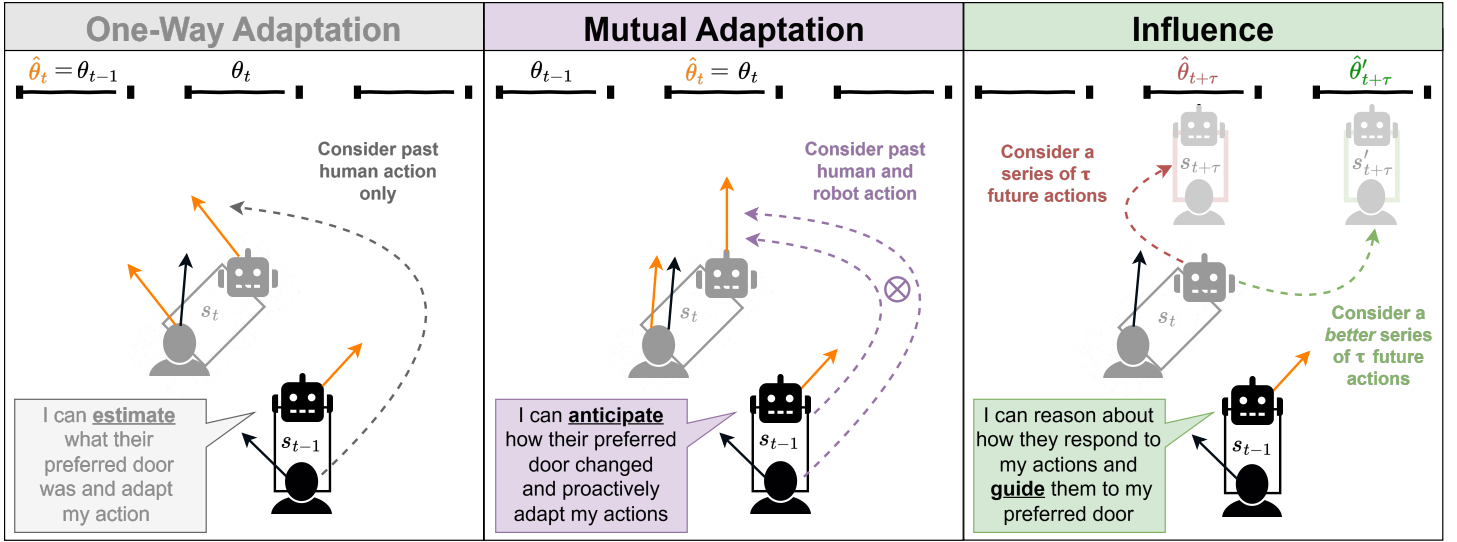


Figure 1: A table carrying task (adapted from [6]) where the goal is for the human and robot to move a table through one of three doors by jointly applying interactive forces (actions) to the table. In the previous state s_{t-1} , the human action (black line from human) and robot action (orange line from robot) was observed and caused the human to update their preferred door from θ_{t-1} (left door) to θ_t (middle door) in the upcoming interaction in s_t . The robot can form a prediction $\hat{\theta}_t$ to deduce what it believes the human action will be (orange line from human) in s_t and plan an action that minimizes conflicting forces. In **one-way adaptation**, the robot uses the past human action in s_{t-1} to *estimate* the human's previous door preference $\hat{\theta}_t = \theta_{t-1}$. This is now incorrect as $\theta_t \neq \theta_{t-1}$ which causes the robot to incorrectly deduce human action (misaligned orange and black arrows from human) and produce conflicting interaction forces (misaligned black human arrow and orange robot arrow). In **mutual adaptation**, the robot now reasons about how its past action affects θ_{t-1} to better *anticipate* θ_t . With the correct the prediction $\hat{\theta}_t = \theta_t$, the robot can align its action in s_t (parallel black and orange lines). In **influence**, the robot can repeatedly apply this reasoning during planning to determine how a series of τ actions (red or green arrows) lead to different door preferences ($\theta_{t+\tau}$ or $\theta'_{t+\tau}$). The robot can then *guide* the human to a latent state that robot better optimizes its reward (i.e., $\hat{\theta}'_{t+\tau}$).

tain a consistent discussion throughout the remainder of the paper. Section 3 describes our proposed latent dynamics framework that unifies the mutual adaptation and influence models for HRI. We then begin the review of relevant works where we state the methodology in Section 4. In the subsequent sections, we review Bayesian (Section 5), Markovian (Section 6), and encoded (Section 7) latent dynamics classes where we first provide formalism for each class and then review how they are used for mutual adaptation or influence in the field of HRI. Section 9 then discusses emergent themes identified in these works and highlights future directions and challenges.

2 Preliminaries

This section provides a brief overview of the concepts and notation underlying the proposed unified framework, as described in Section 3. In order to effectively coordinate, it is useful for the robot to have a model of human policy π_H to predict the human's actions. However, the large dimensionality of π_H can make learning a robotic policy π_R intractable when conditioned directly on π_H . Therefore, a more compact representation of π_H can be used instead, where each unique π_H is described by a latent state $\theta \in \Theta$. The robotic planning problem can then be decomposed into

two steps: 1) **prediction**: the robot attempts to predict what θ can best be used to describe human policy and 2) **control**: the robot attempts to find an optimal policy π_R^* in response to θ such that it optimizes its own reward.

2.1 Interactive Setting

We consider HRI to evolve according to a transition model $s_t \sim P(s_{t-1}, a_{t-1}^R, a_{t-1}^H)$ where the state of the world at any timestep s_t is a function of the previous state s_{t-1} and previous actions a_{t-1}^R and a_{t-1}^H of the robot and human, respectively. Agent i 's actions a_t^i are the output of their policy $\pi_i(s_t, \theta_t)$ which is a function of state s_t and latent state θ_t . After the joint action $\{a_t^R, a_t^H\}$ is performed, the timestep is progressed $t \leftarrow t + 1$ and an observation of the states and actions $o_{t-1} \leftarrow \{s_{t-1}, a_{t-1}^R, a_{t-1}^H\}$ is taken.

2.1.1 Interaction Data

The model for predicting latent state takes the observation from the previous timestep o_{t-1} to make a prediction of the current human latent state $\hat{\theta}_t$ in the upcoming interaction. These observations can be a one-step observation like $\{s_t, a_t^R, a_t^H\}$, a trajectory of observations, or any arbitrary dataset. Depending on the latent dynamic function, ϕ , dif-

ferent types of observations can be used. For example [6] takes a one-step observation to update the latent state, [12] takes the trajectory from the previous interaction, and [13] takes a dataset to update their latent state. For the sake of generality, we use o_t to represent any input in these forms.

2.1.2 Prediction and Control Framework

This framework first predicts the human's latent state and then performs control optimization over a policy conditioned on this prediction. The predictor takes in the previous observation o_{t-1} and predicts the latent state $\hat{\theta}_t$ for the current timestep. Then, the control block finds an optimal robot policy π_R^* which maximizes the expected discounted accumulated reward conditioned on the robot's estimation of human latent state $\hat{\theta}_t$ such that:

$$\pi_R^* = \arg \max_{\pi_R} \mathbb{E} \left[\sum_{\tau=1}^L \gamma^\tau R(s_{t+\tau}, \pi_R, \pi_H(\hat{\theta}_t)) \right] \quad (1)$$

where $\tau \in [1, L]$ is the time index for the unseen future interactions during policy optimization, γ is a discounting factor, and $R(\cdot)$ is some reward function. Here, the time horizon L differentiates the planning time index τ from the time index of the current interactive episode t . Note that the policy optimization is conditioned on the latent state $\hat{\theta}_t$ which is formulated differently between one-way adaptation, mutual adaptation, and influence as shown in Figure 2. Furthermore, when the latent state update takes place (i.e. between or within an interactive episode), determines how frequently (1) is used to solve for π_R^* . We further discuss the latent state in Section 2.2 and Section 2.3.

2.2 Latent State Representation

In a classical ToM setting, latent states represent an agent's mental state, such as intent, beliefs, and desires [2]. Judgments about the other's latent states dictate how we understand and remember others' actions, how we respond, and what we predict about their future actions [14]. This concept of latent states has also been expanded into computer science as feature embeddings, which capture the essential features of input data in a lower-dimension space [15].

Latent state representations can be either *explicit* or *implicit* [16]. Explicit representations describe latent states that have a practical description within the task (e.g., navigation goal [17], reach objective [18], and sorting preferences [17]) or about an agent (e.g., adaptability [19], trust [20], and knowledge [21]). Implicit representations describe latent states that have no semantic meaning and are behavioral encodings that maximize predictive accuracy. While some implicit features may encode semantically meaningful states, we do not consider them as explicit unless the values themselves have semantic meaning. In the reviewed works, implicit representations are mostly seen in autoencoding frameworks such as [22].

2.3 Latent State Dynamics

Latent state dynamics assert that the human's latent state *evolves* according to some functional $\phi(\cdot)$ such that $\hat{\theta}_t \leftarrow \phi(\hat{\theta}_{t-1}, \cdot)$. We define the three *latent dynamics classes* that describe how the robot models this update based on the interaction data. While some conceptual overlaps may exist between these classes, this review focuses on the exact formulation and implementation of a model, instead of the type of system that a model predicts, to remove ambiguity. Note that while some works choose to define the bi-directional nature of mutual adaptation as the human being adaptive to some robot latent state (e.g., robot intent [18]), we generalize this in our model and assume that the robot's latent state is implicitly encoded from o_{t-1} in the human latent update ϕ without explicitly defining it (i.e. we skip the step where the human first estimates robot latent state and directly assert that the human updates their latent state from observations). The proposed framework could easily be modified to reflect this.

3 Unified Framework

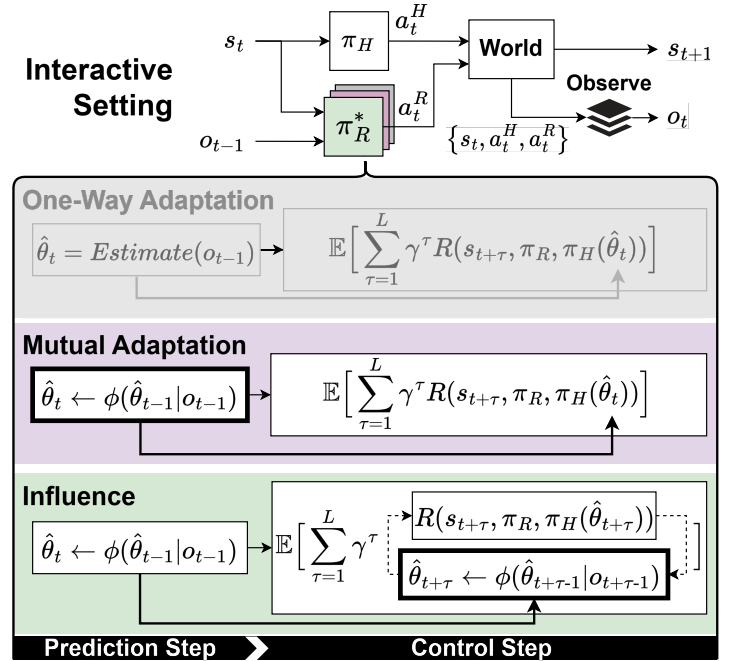


Figure 2: The proposed prediction and control framework showing how one-way adaptation extends to mutual adaptation and mutual adaptation to influence. Changes to the model from the preceding approach are shown by the bolded boxes. The difference between one-way and mutual adaptation is that the latent state prediction of $\hat{\theta}_t = \theta_{t-1}$ from $Estimate(\cdot)$ is improved to the more accurate prediction $\hat{\theta}_t = \theta_t$ by using the latent dynamics function $\phi(\cdot)$ during the prediction step. Mutual adaptation extends to influence by now incorporating $\phi(\cdot)$ into policy optimization during the control step.

In this section, we introduce the proposed framework by

first formalizing one-way adaptation. Then, we show how mutual adaptation is a natural extension by considering that the human's latent state evolves according to a latent dynamics model during the prediction step. Furthermore, we show how influence is an extension of mutual adaptation by considering latent dynamics during the control step. In short, both mutual adaptation and influence utilize latent dynamics, but mutual adaptation focuses on improving one-step predictions of the current latent state, while influence recursively incorporates these predictions into planning in an attempt to guide the human to a more desirable future latent state.

3.1 One-Way Adaptation

Substantial research has been done on one-way robot adaptation to humans. Conceptually, “one-way” refers to the unidirectional adaptation of the robot to the human (i.e. even though they may do so, the human is not modeled as adapting their strategy in response to robot actions). Estimating the human's desired direction of motion from previous state velocities [23] and task-level plans from previously applied forces [24] would be considered one-way adaptation as they only consider previous state or human action while neglecting the effect that the robot action had on the current latent state.

During the *prediction* step in one-way adaptation, the latent state of the human is *estimated* online based on previous observations as in (2).

$$\hat{\theta}_t = \text{Estimate}(o_{t-1}) \quad (2)$$

where the $\text{Estimate}(\cdot)$ function can be any estimation process (e.g., maximum likelihood estimation [25]).

Consider the standard one-way Bayesian estimate of latent state: *the robot* models its belief over human features $b_t^R(\theta)$, which is a *robotic reasoning mechanism* (inference). In contrast, modeling *the human* as performing a Bayesian estimate over robot features is considered a latent dynamics model, since it is instead modeling a human reasoning mechanism. The key distinguishing factor here is where the Bayesian update is located: in the robot's mind (modeling the human) or in the human's mind (modeling the world and/or robot). Thus, simply having the robot attempt to find which θ is most likely to be true (i.e. best describes o_{t-1}) without modeling how the human arrived at θ_t implies that (2) does not satisfy the requirements for the aforementioned latent dynamics classes.

An important characteristic of one-way adaptation is that, without a model of latent dynamics, the best prediction of the human's latent state is the one that describes previous interactions such that $\hat{\theta}_t = \theta_{t-1}$. In other words, there is a one-step time delay in the prediction step for one-way adaptation, causing an inaccurate prediction $\hat{\theta}_t \neq \theta_t$ when the human is sufficiently non-stationary (i.e. the human agent adapts to robot actions and changes their latent state between timesteps). As a result, the subsequent control step will attempt to solve (1) to find a robot policy conditioned

on an outdated prediction of the latent state which results in a suboptimal strategy.

Referring back to our toy example in Figure 1, in one-way adaptation, although the robot tries to estimate the human's preferred door based on the previous observations, it does not consider that this door could have changed since the last interaction due to the human adapting to robot actions. Consequently, this would cause the robot to inaccurately predict human actions and plan poorly coordinated interaction forces.

3.2 Mutual Adaptation

For a robot to adapt to a non-stationary human partner, it may not be sufficient to *estimate* what their latent state was in the previous interaction. Therefore, it can be crucial to *anticipate* what their latent state will be in the current interaction to predict how their behavior changed since the last observation. In other words, it may be necessary to consider human adaptation in response to robot actions when planning robot behavior.

Mutual adaptation occurs when both agents update their control policies in relation to their partner's behavior to fulfill team or individual goals [26]. For the purpose of this review, we define mutual adaptation as *leveraging the human's latent dynamics to better predict (anticipate) the unobserved latent state in the upcoming interaction, thus allowing the robot to adapt its control algorithm accordingly*. This complies with other definitions of mutual adaptation [6, 26, 27].

While mutual adaptation has been formulated as an emergent feature of one-way adaptation as in [28], we only use the term mutual adaptation when the robot is benefiting from implementing latent dynamics in the *prediction step only*, as seen by the bolded box in mutual adaptation's prediction step in Figure 2. Conversely, the use of latent dynamics in the control step would instead satisfy our later definition for influence in Sec. 3.3.

In mutual adaptation models, it is assumed that the human's latent state θ_t changes according to a latent dynamics function $\phi(\cdot)$ in response to the previous observations o_{t-1} containing, in part, past robot actions. Having access to ϕ will allow *anticipation* of the human's latent state and their policy at the current timestep, consequently solving the aforementioned issue of one-way adaptation (i.e. $\hat{\theta}_t \approx \theta_{t-1}$). A general way to express the latent update $\phi(\cdot)$ is:

$$\hat{\theta}_t \leftarrow \phi(\hat{\theta}_{t-1} \mid o_{t-1}) \quad (3)$$

Using the improved prediction in (3), relative to one-way adaptation, to anticipate latent state will then provide a more accurate solution to the policy optimization problem in (1).

Here, $\hat{\theta}_t$ is predicted from (3) at the current timestep t and is fixed during the policy optimization time horizon $\tau \in [1, L]$. Key to mutual adaptation, the *control step* is now

solved over a fixed $\hat{\theta}_t \approx \theta_t$ instead of $\hat{\theta}_t \approx \theta_{t-1}$ as in one-way adaptation.

Returning to our table-carrying example in Figure 1, the human may adapt to previous robotic interaction forces by changing their preferred door from the left door θ_{t-1} to the middle door θ_t . By anticipating this change, the robot can more appropriately adapt its actions to reduce conflicting interaction forces that would otherwise occur when applying one-way adaptation due to the robot's estimate of the human's preferred door being $\hat{\theta}_t \approx \theta_{t-1} \not\approx \theta_t$. As a result, in mutual adaptation, the robot can better respond to the human's adaptive behavior by correctly predicting the middle door as the human's current preference (i.e. $\hat{\theta}_t \approx \theta_t$) to improve coordination.

3.3 Influence

Expanding on mutual adaptation, influence takes the idea of anticipating the human's current latent state a step further by repeatedly applying the same logic during the control step. Thus, we have a model of how the human latent state will evolve given a series of future actions which, in turn, affords the robot the ability to optimize a policy that *guides* the human to a latent state that better supports robot objectives.

For the purpose of this review, we will define influence as *leveraging the human's latent dynamics in robotic planning (control step) to generate observations such that the human is guided to a latent state that better optimizes expected robot reward*. Our definition is most similar to “the ego agent learns to generate interactions...which lead the other agent to adopt future latent strategies that the ego agent can exploit for higher rewards” that is provided in [22], but is generally compliant with descriptions provided in other reviewed works.

Influence has two prerequisites: 1) we have a sufficiently accurate model of the human's latent dynamics and 2) that the robot has the means to take a sequence of actions $\{a_t^R, \dots, a_{t+\tau}^R, \dots, a_{t+L}^R\}$ that generate a set of observations $\{o_t, \dots, o_{t+\tau}, \dots, o_{t+L}\}$ used by the human during their latent state update (i.e. the system is at least partially *control-lable*).

Formally, we will describe this by extending the mutual adaptation model in Section 3.2 to arrive at our model of policy optimization for influence:

$$\pi_R^* = \arg \max_{\pi_R} \mathbb{E} \left[\sum_{\tau=1}^L \gamma^\tau R(s_{t+\tau}, \pi_R, \pi_H(\hat{\theta}_{t+\tau})) \right] \quad (4)$$

$$\text{s.t. } \hat{\theta}_{t+\tau} \leftarrow \phi(\hat{\theta}_{t+\tau-1} \mid o_{t+\tau-1})$$

Here, the online prediction step (anticipation) is identical to mutual adaptation as described in (3). The key difference between influence and mutual adaptation is that $\hat{\theta}_{t+\tau}$ is now subject to $\phi(\cdot)$ and is time-varying during the control step's policy optimization. This is what gives influence its *emergent* property [22]. The policy optimization in (4) is still optimizing for the robot reward $R(\cdot)$ and only uses the latent

dynamics function as a means to *explore and exploit* the modeled system (i.e. world and latent state dynamics) to better maximize its expected reward. Naturally, the robot objective can be defined to induce a specific latent state for the human as in [29], but this is not required for influence to occur.

In summary, the robot can now repeatedly reason about $\phi(\cdot)$ in the same way that mutual adaptation did to anticipate the current $\hat{\theta}_t$ during influence's control step in (4). Now, recall the example of the collaborative table carrying task in Figure 1. This modification yields the ability to evaluate how a sequence of robotic interaction forces can alter the human's preferred door in the future $\hat{\theta}_{t+\tau}$. Thus, the robot has the opportunity to explore different possible latent state outcomes and “choose” the latent state that best suits its needs. Given that the rightmost door has a higher payoff for the robot, the robot will naturally learn to influence the human's preferred door during policy optimization to realize this superior payoff. For example, the robot may learn to apply large interaction forces to the right early in the interaction to elicit the human to prefer the rightmost door in the future. Thus, this small change of allowing $\hat{\theta}_{t+\tau}$ to evolve according to $\phi(\cdot)$ during policy optimization can achieve a large change in policy.

For the purpose of this review, we only consider a robot to be influential if it explicitly models latent dynamics during policy optimization (4). Therefore, influence must be *intentional* and *modeled* within robotic planning. This excludes more passive means for influence such as an unpowered robot “influencing” the path of a human walking by or, more realistically, methods such as [18] that promote legible motion without modeling how such robot actions shape the human's latent dynamics toward certain states.

3.4 Summary

This paper aims to demonstrate how the latent dynamics function $\phi(\cdot)$ is utilized in various HRI tasks, enabling researchers to implement mutual adaptation or influence within our proposed framework. The distinction between mutual adaptation and influence arises from the way $\phi(\cdot)$ is integrated into the prediction and control steps. It is important to note that the proposed framework is agnostic to the latent dynamics class and the policy solution method used in the control step.

The latent dynamics function can be implemented using a variety of methods, which we refer to as latent dynamics representations. This consists of both traditional control approaches [30] and modern deep reinforcement learning methods used in many of our reviewed works. While we briefly discuss how these approaches incorporate $\phi(\cdot)$, we refer the reader to [31, 32] for a more in-depth exploration of these techniques in robotics and HRI. Additionally, we examine various methods for modeling the latent dynamics function $\phi(\cdot)$, ranging from simple hand-crafted functions to autoencoders, Bayesian networks, recurrent neural networks (RNNs), transformers, and more. However, our primary

focus is on how these latent state dynamics modeling approaches are integrated into different HRI tasks, their role in capturing human behavior, and how they are used for mutual adaptation or influence. Therefore, we consider detailed discussion of exact methods for solving the control step (e.g., deep reinforcement learning algorithms) to be largely out of scope.

4 Review Methodology

For our review purposes we searched Google Scholar and four popular databases: Scopus, IEEE Xplore, Science Direct and Springer Link. We used the search terms of “adaptation” or “influence” in fields of “human-(robot/machine/AI/automation /multi-agent),” with additional keywords “latent,” “intent,” and “latent dynamics” in the article abstract, title, or author specified keywords to target articles with a ToM framework.

As the field has yet to agree upon a precise set of terminology, works may apply latent dynamics models in their algorithms but not refer to them by name. We attempted to mitigate the impact of this by performing an additional breadth-first search of works in the first round of articles meeting our selection criteria.

The selection criteria for articles are as follows:

- Must be applicable to HRI tasks. Application-specific approaches that do not reflect the dynamic nature of HRI, such as multi-robot interaction, were not included.
- Must utilize a physically or virtually embodied robotic agent, able to act in the environment independently (e.g., purely text-based chatbots do not qualify). To allow for socially assistive robots, we do not require direct physical interaction between the human and robot [21].
- Must explicitly use a latent dynamics model for the human within the robotic planning framework.
- Must explicitly use some representation of latent human state, although exact terminology may vary.

After filtering by the selection criteria, we uncovered a total of 51 papers that fulfill the goals of this review. The qualifying works were then evaluated along the following dimensions:

- **Approach:** The focus of this survey is to review two *approaches* to applying latent dynamic models in HRI: *mutual adaptation* and *influence*.
- **Latent Dynamics Class:** Latent dynamics model non-stationary humans with an evolving latent state. In our review, we have identified three distinct *classes* of latent dynamics functions: *Bayesian*, *Markovian*, and *encoded* dynamics. Formal definitions of these

classes can be found in Sections 5, 6 and 7 respectively.

- **Latent Dynamics Representation:** Within each latent dynamics class, we further organize works according to their specific model and implementation, as introduced in Section 3.4. We go deeper and describe a unique set of models and methods for each class.
- **Latent State Representation:** Furthermore, we motivate the use of latent state and its formulation in Section 2.2 where we define two types of *latent state representations*: *implicit* and *explicit*.
- **Update Frequency:** We further discuss papers based on when the update of the latent state is applied with respect to the interaction. Depending on the application, this latent state update can occur *within* a single interactive episode (e.g., at each time step of the interaction) or *between* repeated interactions, where the entire previous interaction episode may influence the human’s latent state in the next one.

The justification for using these dimensions to discuss each **approach** stems from the following rationale: Models of how human latent state evolves fit into one of three distinct **latent dynamics classes** that describe the underlying structure and assumptions of varying methods [33]. However, the exact implementation (i.e., **latent dynamics representation**) can vary within each latent dynamics class and was found to motivate use of certain **latent state representations** [16]. Lastly, we found that the timing of the dynamics **update** yielded significantly different results in real-world applications [34].

For the reader’s reference, a summary of the aforementioned dimensions classifying the reviewed works is provided in Table 1. A more detailed discussion of these dimensions will then be provided in Sections 5, 6, and 7.

4.1 Limitations

This review focuses specifically on mutual adaptation and influence methods that leverage explicit modeling of latent dynamics. Although outside the scope of this review, there are other methods of achieving directed mutual adaptation and influence: reward manipulation and explicit design of heuristic influence measures [37], explicit announcement of intent [72], information sharing [73], online learning [74], logic-based planning [75], sampling-based or data-driven modeling with latent dynamics [76,77]. The remainder of the paper is organized based on latent dynamics class (Bayesian in Section 5, Markovian in Section 6, and encoded in in Section 7), where we first describe the model by highlighting the strengths and limitations and then review relevant works for that class as it is applied to mutual adaptation and influence.

Table 1: Taxonomy of Latent Dynamics Dimensions

Class	Approach	Dynamics Representation	Update Frequency	Citations
Bayesian	Mutual Adaptation	Belief Update	Within	[35, 36]
	Influence	Belief Update	Between	[37, 38, 39]
		Belief Update	Within	[18, 40, 41, 42]
Markov	Mutual Adaptation	Fixed Function	Between	[43]
		Fixed Function	Within	[30, 44, 45, 46]
		Transition Matrix	Within	[6, 19, 26, 47]
		Beta-Distribution	Between	[48]
	Influence	Beta-Distribution	Within	[49, 50]
		Fixed Function	Between	[51, 52]
		Linear Gaussian System	Between	[53]
		Linear Gaussian System	Within	[20, 54, 55]
Encoded	Mutual Adaptation	Transformer Matrix	Within	[21, 56, 57, 58]
		Autoencoder ^a	Between	[59, 60]
		Autoencoder ^a	Within	[10, 61, 62]
		Fixed Approximator	Within	[11, 63]
		Predictive Network	Within	[64, 65]
		Transformer	Between	[17]
	Influence	Transformer	Within	[66]
		Autoencoder ^a	Between	[13, 22, 29, 67]
		Autoencoder ^a	Within	[68, 69]
		Predictive Network	Between	[70]
		Predictive Network	Within	[71]
		Transformer	Within	[12]

^aautoencoders are the only dynamics representation that afford implicit latent state representation. All other models use an explicit latent state representation.

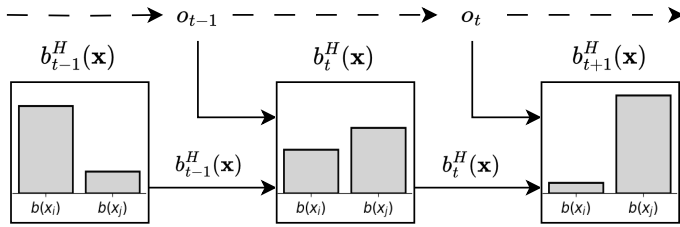


Figure 3: Given a stream of observations o_t , a Bayesian latent dynamics model for the human consists of them updating their belief distribution $b_t^H(\mathbf{x})$ over a set of auxiliary features $\mathbf{x}$ using observations o_{t-1} as evidence in their BeliefUpdate($\cdot$). The robot typically directly adopts the human belief distribution as the current latent state $\hat{\theta}_t := b_t^H(\mathbf{x})$ at every timestep.

5 Bayesian Latent Dynamics

Here, it is assumed that the human maintains a belief $b_t^H(\mathbf{x})$ which is modeled as a probability distribution over a set of features $x_k \in \mathbf{x}$ at time t . These features can describe either some characteristic of the world or their robotic partner where the distribution over $b_t^H(\mathbf{x})$ evolves with t based on some observation o_{t-1} , as seen by the sequences of blocks in Figure 3. Based on o_{t-1} , the agent performs a Bayesian belief update on $b_{t-1}^H(\mathbf{x})$ to form a new belief distribution $b_t^H(\mathbf{x})$ using Bayes rule for each x_k :

$$b_t^H(x_k | o_{t-1}) \propto P(o_{t-1} | x_k) b_{t-1}^H(x_k) \quad (5)$$

From the human’s perspective, given o_{t-1} which consists of the observed states and joint-actions, the belief update at time t is a deterministic function of the prior belief $b_{t-1}^H(\mathbf{x})$. As a shorthand for this deterministic computation, we write:

$$b_t^H \leftarrow \text{BeliefUpdate}(b_{t-1}^H | o_{t-1}). \quad (6)$$

where $\phi := \text{BeliefUpdate}$ for Bayesian latent dynamics. Here, the Bayesian belief update can be carried out with a singular observation from a previous timestep [35] or collection of observed trajectories [78].

An important note discussed in Figure 3 is that the prediction of latent state $\hat{\theta}_t$ is directly expressed as the human’s new belief distribution $b_t^H(\mathbf{x})$ at time t such that $\hat{\theta}_t := b_t^H(\mathbf{x})$. Consequently, the set of possible latent states Θ contains all feasible human belief distributions.

Here, we strictly maintain that the human is a Bayesian learner and is not concerned with robotic reasoning mechanisms (e.g., Bayesian inference of human intent by the robot is not a sufficient condition for Bayesian latent dynamics). Consequently, all Bayesian latent dynamics models assert that the human performs the belief update in (5). We will refer to this latent dynamics representation as *belief update*.

Modeling the human with Bayesian dynamics provides the simplest and most direct of the reviewed models as there is no learning or training involved in the update rule. However, such a simple dynamics model with a fixed procedure comes with some limitations. Computationally, Bayesian models may struggle to be tractable in high-dimensional latent spaces due to the belief on every latent state needing to be updated in each transition. We also observe that the outcome of a transition is highly dependent on prior belief. While a prior can be a powerful feature to integrate pre-existing information, a poor choice of prior can lead to misinterpretation of new evidence and degrade the model. Next, Bayes rule relies on fixed and rational assumptions about the probabilistic relationship between variables which limits its ability to personalize the dynamics to different types of humans. For example, humans have a tendency to resist change in their belief (i.e., confirmation bias [79]) and suboptimally weight or interpret new evidence based on its availability or recency [80]. Similarly, Bayesian latent state dynamics is only relevant when modeling the human’s belief about something other than themselves and is thus limited in the types of latent states it can describe.

5.1 Bayesian Mutual Adaptation

One way to model the human’s latent dynamics in a mutual adaptation setting is by assuming that the human performs a Bayesian update to recover their new belief according to (6). The examples of using the Bayesian latent dynamics model in mutual adaptation scenarios can be found in [35, 36] where both articles use an explicit representation of the human’s latent state and update their latent dynamics model within interactive episodes.

In [36], in a multi-agent interaction scenario, the “intimacy of the partner” has been used as an explicit latent state, which refers to the human agent’s friendliness toward the ego agent in a game-theoretic framework. A Bayesian network was utilized and trained using expectation maximization of the observations to update a friendliness parameter. The approach’s effectiveness was examined in a mutual adaptation task (an iterative prisoner’s dilemma game) by

showing how the predicted friendliness from the latent dynamics model increases with each iteration as the agents get to know each other.

Similar approaches have also been used for human intent recognition tasks [35] where the human intention in an HRI task has been modeled as a sequence of latent states of an evolving hidden Markov model which was first trained on observed data using Baum-Welch algorithm. The human intention was recognized as the most likely latent state sequence based on the output of the Bayesian update using the recent observations. The success of the approach was shown by the positive self-report of the participants in a collaborative assembly task with the robot.

5.2 Bayesian Influence

Applied repeatedly during planning in the control step, the robot can find a way to optimally influence human belief such that the series of beliefs $b_{t+\tau}^H(\mathbf{x})$, $\forall \tau \in [1, L]$ better allows the robot to optimize its reward over some time horizon. This mechanism of planning over the human's inference is often referred to as *inference inversion* [41].

We refer to this process as Bayesian influence, which is typically used when the human is updating their belief about a latent state describing their robotic partner. This can take the form of the human's belief over robot goals [18, 38, 42], reward [37, 39], or internal state [40, 41]. We observe that all Bayesian influence approaches model dynamics over an explicit latent state, as human belief must be contextualized within a representative feature. Generally, the goal of Bayesian influence is to better align human belief with reality [70], increase human understanding of robot actions, or improve team performance [18, 38, 39, 42]. However, it can also be done to promote certain perceptions of robot characteristics, such as attention [41] or trust [40]. Of particular note is the concept of legible motion, where the robot seeks to generate motion that allows a Bayesian human to better infer the robot's navigation goal [18, 38]. Here, successful influence can be verified by showing more accurate human prediction of robot actions or by showing the corresponding latent state change through self-report measures (e.g., increased trust from post-test surveys [40]). Alternatively, in [37], the authors take the opposite objective and attempt to increase uncertainty in the human's belief of robot goals between interactions by injecting entropy into robot actions. This was done to better maintain long-term influence between interactions, which would otherwise degrade once the human learns to anticipate the robot's actions. These authors verified that their approach better maintained long-term influence in an intersection-type task by being able to influence the human to insist on going first less frequently.

Influencing human belief about the robot, particularly about robot goals, is likely more popular than attempting to influence beliefs about the world due to a greater ability to control latent state with robot actions and its compliance with a ToM framework. However, influencing a Bayesian human's belief about the external world is shown in [70],

where a robot attempts to provide modified observations to a Bayesian human to better align their belief about the world state with the true world state in a driving task such that humans could be influenced towards more optimal behaviors.

6 Markovian Latent Dynamics

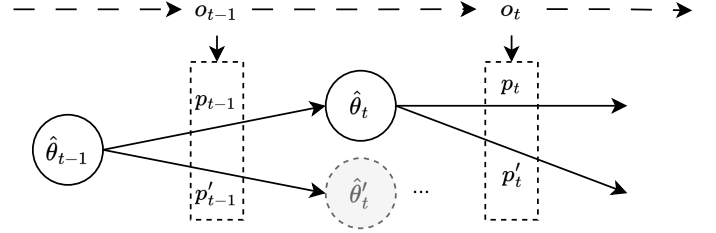


Figure 4: Given a stream of observations o_t , a Markovian latent dynamics model consists of the human stochastically transitioning between singular latent states $\hat{\theta}_t$ with probability p_t determined by their observation o_t . Here the human only follows a single process and occupies one latent state at a time in reality but the robot can probabilistically reason about which process has occurred to form a belief distribution over Θ .

Markovian latent dynamics assert that the human's latent update ϕ maintains the Markovian property: the latent state θ_t depends only on the previous latent state θ_{t-1} and some observation o_{t-1} . Hence the dynamics of a latent Markovian system can be expressed as:

$$\hat{\theta}_t \leftarrow f(\hat{\theta}_{t-1} \mid o_{t-1}) \quad (7)$$

where $\phi := f$ for Markovian latent dynamics models.

Here, the human is assumed to occupy a *single* unknown latent state at any time, and to transition between discrete latent states in a stochastic manner. In other words, the human can only follow a single Markovian process, as shown in Figure 4.

Again, this classification only applies to the human latent dynamics model and not to robotic reasoning mechanisms. For example, Bayesian inference methods can estimate parameters that specify a Markovian human transition function (p_t in Figure 4) or reason probabilistically about what Markovian process the human followed. These methods would maintain the Markovian latent dynamics classification as they still assert that the human follows a Markovian transition, albeit with some uncertainty about exactly which process is being followed.

These transitions in Markovian dynamics can be either stochastic or deterministic. To further clarify, we use the term “Markov” to describe the memoryless quality of these models; hence, while Figure 4 depicts a stochastic process, first-order deterministic models like [30] can still be classified under Markovian dynamics in this review.

While all Markovian latent dynamics can be expressed as a transition matrix, we will focus on organizing them according to methods for recovering transition probabilities. First, this can be done by directly learning transition matrix entries from data with methods like the Baum-Welch algorithm [81]. We will refer to this latent dynamics representation as a *transition matrix*. Alternatively, one may choose to specify transition probabilities to follow a Markovian function that constrains these transition probabilities. These include *beta-distribution*, *linear Gaussian system*, or *fixed functions*. The latter refers to custom-designed functions that do not learn or fit model parameters from data.

Once a transition function is defined, Markovian latent dynamics are the most computationally efficient of the reviewed latent update procedures. However, the inherent assumption in all types of Markovian latent dynamics is that the system cannot account for the history of the latent state (i.e., memoryless) and thus cannot model long-term dependencies. Depending on the latent state, this may be a limiting factor as there may be many latent systems in real-world HRI that are not Markovian (e.g., fatigue or mood does not depend solely on your state one minute ago). Most commonly, Markovian latent dynamics are specified by some constraining function that is either manually authored (i.e., fixed functions) or has learnable parameters. These approaches require a degree of prior knowledge about the latent system such that the constraining function's assumptions are not invalidated. However, functions with learnable parameters (e.g., beta distribution [48, 49, 50]) may be more flexible in describing real humans but require a modest dataset to learn from. Without some prior knowledge, Markovian dynamics may instead solely rely on learning transition probabilities from data (e.g., Baum-Welch algorithm). However, this comes at the cost of a higher data requirement, such that every latent state transition in the system is observed a sufficient number of times to define a probability. Thus, there are a spectrum of approaches to Markovian latent dynamics that balance the trade-off between limitations of prior knowledge and available training data.

6.1 Markovian Mutual Adaptation

Markovian latent dynamics models focus on anticipating the current latent state $\hat{\theta}_t$ directly based on the previous latent state $\hat{\theta}_{t-1}$ according to (7). The key feature of these models is their Markovian property, which makes them a convenient choice for mutual adaptation tasks as they rely on predicting the next latent state only based on the current time information.

Two approaches to representing the latent dynamics function were observed among the reviewed Markovian mutual adaptation models. The first approach relies on the task-specific latent dynamics representation that uses high-level information regarding the task (e.g., trust models) [43, 44]. We will call these Markovian models the “fixed function” dynamic representation. The second approach is to construct the latent dynamics in the form of a transition matrix

that encodes the probability of transition from the current latent state to the next one [6, 19, 26, 47].

Many works in physical HRI (pHRI) have utilized a game-theoretic framework to construct a deterministic Markovian latent dynamic representation (e.g., a first-order dynamic) [30, 45, 46]. The importance of relying on a deterministic latent dynamic in the mentioned works is that it allows for the application of optimal control-based approaches to demonstrate performance guarantee (e.g., Lyapunov stability) or to formulate update rules for modeling human behavior (e.g., through coupled Riccati equations [30]). Based on the dynamic game's coupled nature and the human agent's optimality assumptions, [30, 45, 46] have successfully generated latent dynamics functions that update the latent state within interactions. In this framework, the latent state is represented explicitly as either the human's control gains during the physical task [30] or the amount of human contribution in a cooperative game played between the human and the robot. Based on the reviewed works [30, 45, 46], this game theoretic interactive framework allows for representing the latent dynamics in different task-specific ways. In [30], the authors modeled a coupled physical interaction using a dynamic game by considering the human and the robot as two players, each minimizing their corresponding cost function. This formulation resulted in two coupled Riccati equations for each player; then, they proposed an update rule to represent the latent dynamics for predicting the human matrix solution to their Riccati equation within interactions and consequently anticipated the human's control gain.

In [43], another game-theoretic formulation has been used to model the human as a partially adaptable agent to the robot in a table-clearing task. It was assumed that the human was unaware of the robot's payoff function entities at the beginning of the task and might learn those entities after each set of interactions with a probability of α and might not learn it with a probability of $1 - \alpha$ (partially adaptable). The human understanding of the robot's payoff entities can be understood as the latent state, and the probability function of learning the entities can be understood as the latent dynamics in this framework. Compared to the previously mentioned game-theoretic approach that updates the latent state within interactions, the latent state update in this work occurs between each set of interactions.

The above-proposed models were validated by their success in predicting human behavior [43] or by improving the overall human-robot performance [30, 45]. However, game theoretic or optimal control formulations of HRI often rely on strong assumptions about human rationality or full knowledge of cost functions which may be limiting, but also afford analysis of convergence guarantees as in [30].

To address the above-mentioned problems, partially observable Markov decision processes (POMDPs) [82] and mixed observability Markov decision processes (MOMDPs) [83] have been used to reason about uncertainties regarding human intentions [84]. In [6, 19, 26], a MOMDP framework has been used to model the mutual adaptation task and structure the latent dynamics setting. In [6, 19, 26], the authors have introduced the bounded-memory adaptation model

(BAM) to capture human adaptive behaviors. The BAM assumes that the human operates in one of several collaboration “modes” and adapts the behavior by switching among the modes based on their decision to adapt to the robot or resist the robot’s decision. The human adaptability level, which is the probability of switching from one collaborative mode to another, is a BAM model parameter unknown to the robot a priori. Basically, BAM represents the latent dynamics as a transition matrix that captures human adaptability and predicts the next latent state (the collaboration mode) within interactions based on the current latent state and the adaptability level. In [26], the BAM model was integrated into a POMDP. However, MOMDPs have been shown to have high computational efficiency in motion planning tasks compared to POMDPs [26, 85].

Another use of transition matrices for representing human latent dynamics in a mutual adaptation task can be seen in [47], where human intent is considered as an explicit latent state. The authors of this work provide an intent prediction framework using transition matrix modeling of the latent dynamics to update the latent state prediction within interactions. The authors proposed a Bayesian generative model for social navigation that generalizes a popular class of deterministic multi-agent motion models (in navigation) and is capable of encoding traffic agents in probabilistic space. Their model learning was formulated as a maximum likelihood estimation problem, maximizing the marginal likelihood by integrating out the latent dynamical states from the joint probability of the complete data (observed and latent). Notably, they showed that their algorithm’s inference run time scaled linearly with task complexity.

Modeling human trust as an explicit latent state in mutual adaptation tasks is also a common way of anticipating human behavior during interactions [19, 26, 44]. An example of using trust models as a Markovian latent dynamics representation in mutual adaptation can be seen in [44] where, inspired by the time series human-to-robot trust model [86], a fixed function dynamics model for predicting human-robot mutual trust has been proposed. This model was used in a (semi)autonomous underwater robot control task where the robot updated the trust parameter prediction within interactions and decided whether to switch between the teleoperated and autonomous mode to fulfill the objective in a mutual adaptation task with the human operator.

6.2 Markovian Influence

Next, we will look at how existing works use Markovian latent dynamics to influence the human. It is important to keep in mind that the robot can maintain Bayesian reasoning over a human modeled with Markovian latent dynamics, as is the case in Figure 4, and that we are only concerned with the latter to classify Markovian latent dynamics functions. This is particularly relevant in POMDP settings as the structure prompts this paradigm.

Of particular note is that Markovian latent dynamics models are most commonly applied to influence human trust.

This is because trust is “significantly influenced by trust at the previous moment” [86], which satisfies the Markovian property, and is critical for maintaining effective interaction online by preventing over or under-reliance [87]. The latter criteria implies that trust would benefit from actively modulating (influencing) trust, not just mutually adapting to it. The trust-influence problem is applied so that the robot can modulate human trust to align it appropriately with robotic capabilities.

In the reviewed trust-influence works, a trust dynamics model is typically used to constrain the full Markovian dynamics. The authors in [51, 52] used a fixed function based on the explicability of robot actions. Alternatively, one may specify a constrained transition function with a model with learned parameters. An example of this is using the Beta-distribution to model trust evolution where model parameters are recovered using a gradient decent algorithm either between [48] or within [49, 50] an interaction. A linear Gaussian system trust model was also used in [20, 53, 54, 55] where model parameters were recovered using Bayesian inference using Hamiltonian Monte Carlo (HMC) sampling [88]. All of these trust dynamics models apply their respective constraints by informing the Markovian system of how trust is impacted by robot capability or other relevant features [86]. A parameterized Markovian function is commonly implemented due to the balance between the integration of prior trust evolution information that constrains the Markovian transition and the personalization provided from learning model parameters. However, one work did choose to ambiguously learn the trust transition matrix directly from data (i.e., no constraining trust model) [57] using an extended Baum-Welch algorithm [81]. In all cases except for fixed function approaches, a data prior may be a limiting in such settings as collecting labeled transitions requires costly interventions like trust surveys.

This trust-influence problem is often solved as a trust-MDP [49, 50, 51, 52, 57] which asserts that trust is the state of the system. Similarly, [20, 55] proposed the trust-POMDP model where trust dynamics are modeled as a linear Gaussian system and is nested within the cumulative POMDP transition dynamics. Solving the trust-POMDP under this trust dynamics model allows the robot to modulate appropriate trust to achieve effective interaction as in [53], where the authors used this framework for trust-based navigation of autonomous vehicles where trust dynamics are modeled between interactions. In [54], the authors extend this linear Gaussian system in the trust-POMDP model by implementing a sparse Gaussian process to address dynamic robot capabilities to improve accuracy in the trust evolution model within an interaction. Modeling trust is a popular topic of research, and there are several other models of trust dynamics (see [89] for a review of potential models) that have yet to be applied to solve this trust-influence problem. Typically, this problem is shown to successfully demonstrate influence by verification of appropriate trust levels through self-report surveys post-test since this is the only way to recover a ground-truth measure of this latent state.

Similarly to trust, Markovian influence has been applied

to other human factors objectives: enhancing human attention [56] and knowledge [21] in POMDP settings where latent transitions matrices are directly learned from data. The former applied the Baum-Welch algorithm and successfully observed influence through improved response times while the latter directly observed the likelihood of transition probabilities from prior data and showed the ability to influence through post-test surveys. Besides this, Markovian influence can attempt to alter human goals such as reach objective [58] by applying an Agent Markov Model (AMM) [90] that specifies latent transitions within agent decision dynamics. Interestingly, these Markovian influence methods learn unconstrained transition functions directly from data, likely because pre-existing models for these latent state transitions do not yet exist since they are often task-specific. This is not the case with trust, where trust dynamics are often determinate of features common throughout all tasks (e.g., performance), thereby allowing ubiquitous models to be available.

Markovian influence tends to lend itself to modeling and influencing humans within a single interaction [20, 21, 49, 50, 54, 55, 56, 57, 58]. This is due to the form of the latent dynamics being explicitly defined for each state-observation pair which is particularly compliant with updating within interactive episodes. However, Markovian influence can also take place between interaction episodes if framed as a meta-MDP problem [48, 51, 52, 53] which plans to influence hierarchically separate from low-level planning. Also, similarly to Bayesian influence, all the reviewed Markovian influence approaches model an explicit latent state.

7 Encoded Latent Dynamics

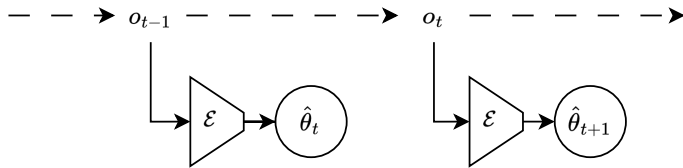


Figure 5: Given a stream of observations o_t , an encoded latent dynamics model for the human consists of encoder using a less compact representation of latent state o_t to predict the next latent state $\hat{\theta}_{t+1}$. Here, the “dynamics” is captured by a time differential between the input and output of the encoder where, typically, the robot discretely adopts the encoded latent state instead of forming a belief over Θ .

In encoded latent dynamics, we do not directly find a map from the previous latent state to the next. Instead, we rely on less-compact representations of human latent state o_{t-1} that implicitly encode $\hat{\theta}_{t-1}$ to predict $\hat{\theta}_t$. Encoded latent dynamics therefore use an encoder $\phi := \mathcal{E}$ as our latent dynamics function:

$$\hat{\theta}_t \leftarrow \mathcal{E}(o_{t-1}) \quad (8)$$

Notice how the *current* latent state $\hat{\theta}_t$ is determined by the *past* observation o_{t-1} , which characterizes the *past* latent

state $\hat{\theta}_{t-1}$, thus giving us a function expressing dynamics. This can be seen in Figure 5 where each prediction of $\hat{\theta}_t$ is not dependent on the previous prediction while dynamics is still expressed through the time differential between the input o_{t-1} and output $\hat{\theta}_t$ of $\mathcal{E}$. In the reviewed works, encoded latent dynamics representations consist of *autoencoders*, *predictive networks*, *transformers*, and *fixed approximators*. Similar to the aforementioned fixed function, the latter representation refers to encoded approximators with fixed parameters that are not learned.

Encoded latent dynamics methods are typically the most flexible as they can take any arbitrary input to ϕ . However, a large portion of these approaches focus on using deep learning to recover a dynamics function from a dataset in a way that may be susceptible to overfitting. This can be mitigated with a sufficiently large training dataset which results in the main limitation of encoded methods: a sufficiently large dataset may be expensive or infeasible to collect. The computational resources needed to train these models is often larger than those of other methods. Alternatively, in the case of a fixed approximator, these dynamic functions have to be manually authored which can lead to errors if the latent system is not correctly or sufficiently specified.

7.1 Encoded Mutual Adaptation

Encoded latent dynamics methods are another approach to modeling human adaptive behavior and latent dynamics inspired by traditional approaches in robots learning from human data [91, 92]. In contrast to Markovian models and Bayesian models, the encoded latent dynamics model directly maps the stream of observations o_t to $\hat{\theta}_{t+1}$ as shown in Figure 5.

Similar to Markovian latent dynamics models, encoded models have also been used in a game theoretic interactive setting for modeling HRI [11, 60, 66]. Authors in [11] modeled a pHRI task as a cooperative game between the human and the robot and used bargaining theory to update the weight of each player’s cost in the overall cost as the latent state within interactions. The latent dynamic used in this work has a predefined structure based on bargaining theory that gets the previously observed data and outputs the weight of the human’s cost in the overall cost function of the cooperative game. Here, the authors also provide a convergence guarantee of their approach. In [60], the HRI task has been considered as a cooperative Markov game. A neural network called a co-policy has been trained using a dataset of human-human interactions to output the solution of this Markov game at each timestep by getting the current states and the human strategy code as the latent state. Instead of solving for the co-policy using RL, which requires a complex reward function for human-robot collaboration, imitation learning directly learns from demonstrations. Thus, human-human demonstration data is recorded, and co-policy learning is formulated using MA-GAIL, a multi-agent extension of the Generative Adversarial Imitation Learning (GAIL). The implicit latent state is modeled as the output of an autoencoder (as the representation of the latent dynamics)

that gets an observation history and encodes it to $\hat{\theta}_t$. Combining these two trained networks, the robot can predict the strategy code and use it to find the appropriate action using the co-policy network to satisfy the game solution. In another game-theoretic approach [66], a more state-of-the-art representation of encoded latent dynamics is used. The authors have used pre-trained Motion Transformers (MTR) [93], a transformer-based encoder-decoder architecture that takes as input the previously observed trajectory of the human agent and outputs the parameters of a Gaussian Mixture Model (GMM). This GMM is then used to predict the possible future trajectories of the human driver. These predictions are subsequently integrated into a game-theoretic Hamilton-Jacobi-Bellman (HJB) reachability analysis to enable safe trajectory planning for the autonomous agent.

Among all the latent dynamics representations, autoencoders are a popular choice in encoding approaches. They learn features and provide a compact representation of human behavior during tasks as an *implicit* representation of the latent state [10, 59, 61, 62]. The study in [59] focuses on learning effective latent representations for the demonstration of human policies. Using simulation data, the authors trained an autoencoder to predict the upcoming human policy, then adjusted the representation with test-time interaction data to not rely solely on simulated populations. Their model updates the implicit latent state between interactions by maximizing the likelihood of observed human behavior, enabling more accurate human policy prediction for robust robot adaptation in physical assistance scenarios. Alternatively, in [10], the latent state is represented as the human's goal position that does not come directly from an autoencoder. Instead, it is derived from a GMM. In this model, within each interaction, the distribution parameters of the GMM are updated and come from an autoencoder as the latent dynamics representation. A Modified Extended Kalman Filter (MEKF) algorithm fine-tunes the model parameters and executes online adaptation for better individual customization and scenario transfer where they show scalability of their algorithm to different task complexities. Similarly, in [61], autoencoders are used to generate a latent embedding of the human's future joint positions in a feeding assistance task. The predicted joint positions are then used to derive a human-aware policy for the robot, which generates an action that more closely aligns with the user's motion tendencies. Finally, in [62], autoencoders are employed in a pursuer-evader game. Vision-based RGB camera data are used to generate a latent embedding of the human evader's navigation intent via an autoencoder. This latent representation is then used to model the evader's policy, enabling a robotic pursuer to effectively intercept the evading human. The autoencoder is trained offline with fully observable possible evader policies, then the trained network has been used in real time where the human policy is not observable.

The use of predictive networks to represent latent dynamics in mutual adaptation includes the use of RNNs to represent how time series of past observations might affect an explicit latent state. In [65], the latent state is defined as

a compact representation of the physical human state, and the latent dynamics consist of a feedback loop, which is a Markovian function, and a feed-forward loop, which is a neural network that acts within interactions based on the observation history, combining a Markovian approach with an encoded approach. However, as noted by the authors, maintaining the Markovian property in such a framework is challenging. In response to this issue, an RNN was employed to estimate the current latent state from a sequence of past observations. Their network was first trained on a pHRI task dataset and then their model was validated by its success in predicting the next physical state.

Using trust as the latent state has also been observed among encoded latent state dynamics approaches. The OPTIMO model presented in [63, 94] benefits from combining Bayesian and encoded approaches by considering a Dynamic Bayesian Network (DBN) as the latent dynamics representation and predicting human trust within interactions. With a trust parameter prediction, the robot can switch to more conservative (less aggressive) control mode to reduce trust losses. Model validation was done by showing how the robot operating under OPTIMO can accurately adapt to the human collaborator's actions and attitudes compared to other approaches.

The use of state-of-the-art architectures, such as transformers, for modeling latent dynamics in mutual adaptation is demonstrated in [17] and [64]. In [17], the authors develop a human intent prediction model that estimates the probabilities of transitioning between latent states (i.e., high-level intents) across interactions. Although the time-varying intent model is initially introduced as a Markovian function, it is estimated through encoded latent dynamics. Their model is implemented using a transformer architecture that takes the history of observed data as input and is trained in simulation across various navigation tasks to capture all possible high-level human intents during the task. To mitigate the risk of incorrect intent prediction, the model actively queries the human when uncertainty is high.

Finally in [64], Generative Pre-trained Transformers (GPTs) and Multilayer Perceptrons (MLPs) have recently been utilized in constructing a neural network representation of the latent dynamics. The authors developed the method (BeTrans) for both long and short-term human goal anticipation as the latent state. Their transformer was trained in two phases: pre-training and fine-tuning. In pre-training, using simulated human behaviors (to avoid the need for large datasets) they formed their model prediction error to form their training loss for learning the weight of the network. In fine-tuning, the model used predicted latent variables along with state and action inputs to forecast the next state and reward. They validated their method by comparing its performance with ORACLE.

7.2 Encoded Influence

In the same way that Bayesian and Markovian mutual adaptation can be extended to influence, so can encoded latent

dynamics by additionally integrating it into planning during the control step.

Much of the work on encoded latent dynamics to enable influence stems from the creative use of autoencoders. Traditionally, autoencoders create implicit latent embeddings by optimizing reconstruction loss between a feature set identically used as input to the encoder and output to the decoder. However, to capture dynamics, an autoencoder can optimize reconstruction loss between a previous observation o_{t-1} and the same feature set in the next timestep o_t . The reviewed works primarily formulate the interaction problem as a series of HiP-MDPs [95] where the latent state is assumed to be fixed within a single interaction and transition between interactions.

This line of work began with the Learning and Influencing Latent Intent (LILI) algorithm [22] that trains an autoencoder to encode the previous trajectory o_{t-1} and reconstruct the next trajectory o_t using a maximum likelihood objective over observed transitions. Here, the output of the encoder is the implicit representation of the human latent strategy, which can be expressed as $\hat{\theta}_t \leftarrow \mathcal{E}(o_{t-1})$. By maximizing expected reward across future interactions, under the dynamics represented by the implicit encoding $\hat{\theta}_t$, a robot can learn to influence the human between interactions. The authors were able to show that the robot could successfully influence another robot's (human-proxy) shooting strategy in an air hockey experiment, but failed to do so with real humans as the influencing robot's policy was found to be too "brittle" when playing against a human with imperfect action emissions.

This framework was extended to Stable Influencing of Latent Intent (SILI) by adding an auxiliary reward that penalizes large distances between θ_{t-1} and θ_t [29]. This enforces pairwise stability on the human's latent state to promote more consistent interactive policies. By doing so, the robotic agent learns to take actions that influence the human between interactions to maintain their current latent state such that $\theta_{t+1} \approx \theta_t$. Generally, stability was achieved in several settings with a simulated human collaborator.

Alternatively, Robustly Influencing Latent Intent (RILI) [67] was developed to consider latent dynamics functions that differ between individual humans. Here, an additional latent hyperparameter that describes different latent dynamics functions is learned with another encoder trained on the history of past latent states $\{\theta_{t-1}, \theta_{t-2}, \dots\}$. This hyperparameter is then used as input alongside the previous trajectories to predict the next latent state in a similar way to LILI to achieve a latent dynamics encoding. This parameter is then inferred between interactions, and planning is done conditionally (i.e., robustly) by considering possible variations over latent dynamics functions and states. A slightly different formulation of RILI was proposed in [13] where authors forgo this hyperparameter and instead rely on past interaction data $\{o_{t-1}, o_{t-2}, \dots\}$ to inform which dynamics function is being used. Thus, they robustly train a single model with several partners such that the encoder can predict multiple types of latent dynamics functions. While most of these models were verified to influence simulated humans, RILI

showed promise for influencing real humans towards more optimal behavior, where LILI and SILI failed to effectively do so outside simulated settings [67].

Most recently, [69] has attempted to relax limitations of LILI, SILI, and RILI on limited complexity in policies, requirement for online data, and the ability to only influence between interactions by showing the ability to influence humans towards more optimal policies. This was done by training an autoencoder to model latent dynamics using a human-human data prior. With this available dynamics function, an agent was then able to learn to influence offline. While [68] does not directly frame the learning problem in an HRI setting, the FURTHER algorithm applies an autoencoder framework to estimate the latent dynamics of the partner agent in a very similar way to LILI, SILI, and RILI.

In [12], the authors deviate from the autoencoder framework and use a transformer encoder to represent a wide range of possible learning dynamics functions. Here, they simultaneously infer the initial latent state $\hat{\theta}_0$ and the latent dynamics approximation parameters using maximum likelihood estimation and gradient-based optimization over previous observations within an interaction. More directly, in [70], the authors used an RNN to encode a sequence of observations into latent states that approximated a human's biased belief update between interactions. Both former articles show the influence towards more optimal human behavior through a more successful policy. Lastly, the authors of [71] extend this to a multi-human setting where they train a robotic policy to exert social influence on a group by using a dynamic leader-follower graph, approximated by a long short-term memory (LSTM) network, where they are able to show that they can influence the team's navigation goals within an interaction.

8 Applications

In this section, we study the distribution of latent dynamics models used for mutual adaptation and influence tasks in various applications in HRI. To achieve better organization, the classification system proposed in [96] was adapted to sort HRI applications into different groups based on the robot task and the human role.

According to the definitions proposed in [96], robot tasks in various applications can be categorized into five groups: *manipulation* (e.g., pick-and-place [43]), *transport* (e.g., autonomous vehicles [10]), *cognitive stimulation* (e.g., tutoring robots [21]), *information exchange* (e.g., recommendation systems [50]), and *physical stimulation*¹ (e.g., motor impairment assistance [59]). Examples of these tasks can be found in Figure 6. Moreover, the role of the human can be classified in six ways: *collaborator* (e.g., table carrying [6]), *operator* (e.g., teleoperation [26]), *supervisor* (e.g., intermittent control [57]), *cooperator* (e.g., leader-follower games [71]), *coactive*² (e.g., collision avoidance [47]) and *competitor* (air

¹Referrers to both physical stimulation and load reduction in [96]

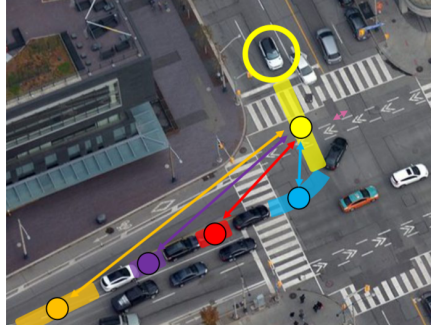
²Referred to as bystander in [96]

Table 2: Taxonomy of Robot Task and Human Role

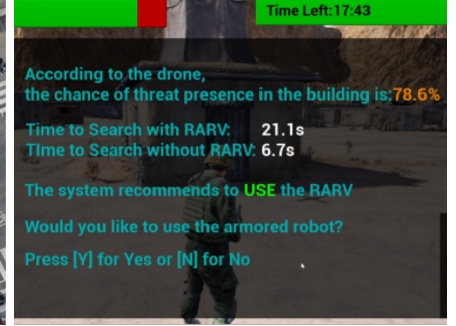
		Bayesian		Markovian		Encoded	
		Mutual Adaptation	Influence	Mutual Adaptation	Influence	Mutual Adaptation	Influence
Robot Task	Manipulation	[35]	[38, 40], [18, 42]	[6, 17, 43], [19, 26, 30]	[20, 55], [57, 58]	[60, 64, 65]	[13, 22, 67], [12, 68]
	Transport	-	[37, 39]	[44, 46, 47]	[51, 52, 54]	[10, 63], [62, 66]	[29, 69, 71]
	Info Exchange	-	[38, 39], [18, 42]	-	[49, 50, 58]	-	[70]
	Cognitive Stim.	[36]	[41]	-	[48, 56], [21, 53]	-	-
	Physical Stim.	-	-	[30, 45]	-	[11, 59, 61]	-
Human Role	Collaborator	[35, 36]	[18, 38]	[6, 26, 43], [30, 45]	[20, 54, 55], [56, 58]	[59, 64, 65], [60, 61]	[13, 67, 69]
	Operator	-	[42]	[19, 44, 46]	[53, 57]	-	[12]
	Supervisor	-	[40]	[17, 44]	[49, 50, 51, 52], [53, 56, 57]	[63]	-
	Cooperator	-	-	[30]	[21, 48]	[11]	[70, 71]
	Coactive	-	[37, 39, 41]	[47]	-	[10, 66]	[29]
	Competitor	-	-	-	-	[62]	[22, 68]



(a) Manipulation with human collaborator [43]



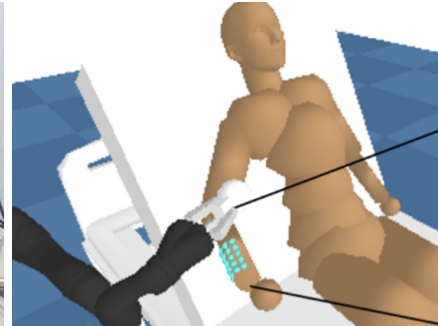
(b) Transport with coactive humans [10]



(c) Info exchange with human supervisor [50]



(d) Cognitive stim. with human cooperator [21]



(e) Physical stim. with human collaborator [59]

Figure 6: Illustration of examples of latent dynamics being used for mutual adaptation and influence in the many domains of HRI. Figures 6a-6e provide a classification of both the robot task and the human role for each cited work.

hockey [22]). Organizing applications in this manner allows for a clear understanding of how different latent dynamics models may be either effective or ineffective in various scenarios. This is particularly valuable when determining whether mutual adaptation or influence is best suited for a particular HRI application. This categorization has been applied to all reviewed literature in this paper and has been summarized in Table 2. For a broader view how these applications correlate with all dimensions presented in this survey, we refer the readers to Figure 7.

An immediate observation from Table 2 is that the majority of the reviewed literature falls under the categories of manipulation (23 papers) and transport (15 papers) in terms of robot tasks. We believe this reflects a natural characteristic of HRI, as current state-of-the-art research in HRI is most commonly framed as a robotic arm or autonomous vehicle interacting with a human. Similarly, a significant portion of the papers consider the human role as a collaborator (21 papers). This aligns with the tendency of HRI to be framed as a collaborative problem, highlighting the strong interest in collaborative tasks within state-of-the-art HRI research.

We can also observe that mutual adaptation saw very few applications in information exchange and cognitive stimulation robot tasks. This is because, according to our proposed framework, these two tasks are inherently an influence problem. In information exchange and cognitive stimulation tasks, as the names suggest, the robot's primary objective is to stimulate or alter the internal state of the human—either by providing information to the human (e.g., legible motion [18, 38, 39, 42], verbal communication [58], text-based communication [49, 50], or synthesized observations [70]) or by influencing cognitive features like knowledge [21], attention [41, 56], or trust [48]. At the same time, we observe that most cognitive stimulation and information exchange tasks utilize Markovian latent dynamics, closely followed by Bayesian latent dynamics, with minimal use of encoded latent dynamic models for these applications. We believe this observation can be attributed to two primary reasons. First, studies have shown that many human internal states are effectively Markovian in nature [97], while Bayesian models are widely regarded as a simple and cognitively valid representation of human reasoning [98]. Second, most encoded latent dynamics models create a unique challenge here as obtaining numerous, high-quality samples to train a dynamics model can be difficult when measuring latent states that can only be observed through costly and indirect means like surveys. However, this only applies to when the model is learned. As observed in Table 2, most applications where the human agent is considered a supervisor employ Markovian latent dynamics, as in such tasks, the robot primarily interacts with human internal states, often with the goal of calibrating trust. Thus, we also observe little works in encoded latent dynamics with a human supervisor due to the aforementioned issue of costly training samples.

In contrast to the limited use of mutual adaptation in cognitive stimulation and information exchange, physical stimulation applications are exclusively proposed as a mutual adaptation problem. This is because these studies focus

on robots providing physical assistance to humans [59] or joint control problems [11, 30, 45] where anticipation of latent state (i.e. mutual adaptation) is sufficient to achieve assistive objectives whereas we typically do not want to be directly changing the underlying features of the human (i.e. influence). While rehabilitation robotics could be a notable physical stimulation task where influence plays a role, this approach has not been explored in existing latent dynamics literature, to the best of our knowledge, likely due to a latent dynamics model of neuromuscular recovery mechanisms being exceedingly complex.

Regarding Bayesian latent dynamics, we observed that it is predominantly used for influencing humans rather than for mutual adaptation. This is likely because Bayesian models primarily deal with the human's belief about external features like the world [70] or, more typically, their robot partner [18, 38, 40, 41, 42]. Thus, Bayesian latent dynamics complies well with robot objectives that seek to align human beliefs with reality, which is inherently an influence problem.

9 Discussion

In this section, we first provide an outline of how researchers may make decisions when implementing latent dynamics. This discussion is provided in the context of the classified works in Table 1 along the listed dimensions of approach, latent dynamics class, latent dynamics representation, and update frequency. Then, we discuss emergent themes that arise from subsets of works to highlight existing trends in the current field of latent dynamics. We support these trends with analysis provided in Figure 7 that shows frequency of co-occurrence between the presented dimensions. Finally, we provide a summary of research areas that could warrant further investigation.

9.1 Latent Dynamics Implementation

Selecting an Approach: The work until this point has been structured to independently review approaches to mutual adaptation and influence within the context of each latent dynamics model. But how do we know which approach is better in practice? Theoretically, influence is a better maximizer of cumulative robot reward since it can not only anticipate, but also leverage sequential reasoning about latent state to its advantage. Putting aside the obvious computational challenge of solving the control step under a dynamic latent state, this does not guarantee its superiority in all practical scenarios. To exemplify this, consider an assistive robot that applies low-limb torques to minimize human walking effort where the human's latent state is their intended gait.

If we have an inaccurate model of how robot torques change the human's intended gait (i.e., latent state dynamics), then the influential actions will not realize the expected higher payoff. In other words, compounding error from latent state

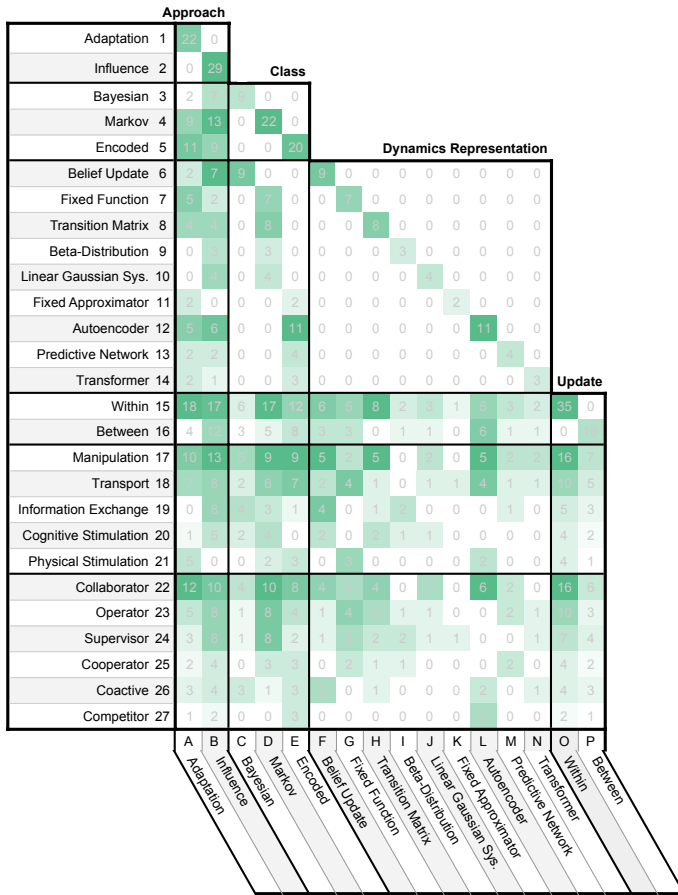


Figure 7: Bibliographic co-occurrence matrix showing the frequency of co-occurrence between all presented dimensions in this paper (defined in Sec. 4 and Sec. 8). Darker colors indicate higher frequencies relative to other co-occurrences within each bounded cross-section of dimensions (e.g., within the 2×3 Update-Class rectangle). Cross-sections may not sum to the total paper count in cases where works satisfied multiple criteria. Co-occurrence between Robot Task and Human Role is excluded for brevity.

dynamics in the control step may produce actions that are largely erroneous. In contrast, this error only occurs once in mutual adaptation during the prediction step. Thus, if inaccuracies are small enough to improve prediction compared to one-way adaptation but generate erroneous influence, then mutual adaptation is likely the best approach. This consideration is especially critical in light of the real-world difficulties in modeling an unobservable transition with noisy observations (e.g., stochastic humans or sensor noise).

Additionally, influence may suffer under an ill-specified objective where the robot function fails to capture all task aspects. Even with a perfect latent dynamics model, if the measured human walking effort (objective) is incorrectly defined or subject to sensing errors, then the control step may produce actions that risk steering the human toward gaits (latent state) with unintended consequences. This effect is less prevalent with mutual adaptation, as the human retains higher task arbitration and is not subject to the erroneous influence.

Finally, influence may not always be ethical to implement [99], especially when this would cause a human to act against their own interests (e.g., develop a compensatory gait that hinders non-assisted walking). Thus, we should only look to influence when we have an accurate model of latent state dynamics, a well specified objective, and it is ethical to do so. Otherwise, mutual adaptation may be a superior approach.

Selecting a Latent Dynamics Class: Once an approach is selected, a latent dynamics class can be chosen depending on access to human data and prior knowledge about the structure of the latent dynamics function as presented in Table 1. The fact that these latent dynamics representations and classes can be used interchangeably within each approach is a direct result of the unifying framework proposed in this paper and provides a structured method for making these decisions. For instance, Bayesian latent dynamics may be useful when there is no human data prior (i.e., the ability to start with a uniform prior) and little additional knowledge of how the human performs their latent update. However, if knowledge of the structure of the latent update is available with some human data to fit a model, a Markovian latent dynamics representation may be best. Finally, encoded latent dynamics are useful for ambiguously learning the update rule (i.e. no prior knowledge of the latent dynamics structure) given that there is access to a sufficiently large human dataset to train the model.

Selecting a Latent Dynamics Representation Similarly to selecting the class, the choice of latent dynamics representation is a matter of personalization and prior knowledge. Fixed functions and approximators both assert a constant representation of the dynamics but alleviate the need for data. Conversely, constraining dynamics to a certain form as in other Markovian representations allows more efficient use of data when learning parameters. Finally, if no prior knowledge of how the latent state progresses is available, one can rely on purely data-driven methods such as encoded representations or learn a Markovian transition matrix directly as in [57].

Selecting an Update Frequency: Rationale for selecting update frequency and exact latent dynamics representation are subject to particular setting of the task. For example, update frequency should be selected based on how often the human latent state is likely to change. For example, within interaction may be better suited for fickle latent states like adaptability [6] while between interaction may be better suited for things like sorting preferences [17] that may have some resilience once the interaction has started.

9.2 Emergent Properties

The remainder of this section will now review particular observations that arose as part of this survey to better allow researchers to make decisions on how to apply these models.

Latent Dynamics Classes: Among all reviewed papers, Markovian models are the most prevalent class for both mutual adaptation and influence (Figure 7, Col A-B Row

3-5). We believe this popularity stems from the usefulness of Markovian dynamics in planning optimal actions [100], their computational efficiency, and capacity to personalize human behavior through parameterization of constraining functions. The latter point comes from the fact that human latent states can often be accurately described as Markovian [97]. For example, the human trust parameter exhibits inherent Markovian properties, as it is significantly influenced by previous moments of trust [86]. In addition, POMDPs are a common interactive setting in the reviewed works and are fully compliant with Markovian latent dynamics due to the partially observable transition function's ability to be decomposed into world (observable) and latent state (unobservable) transition functions.

Conversely, Bayesian latent dynamics are notably less common (Figure 7, Col C). This is particularly surprising given the widespread use of Bayesian inference in one-way adaptation [101, 102, 103], its computational efficiency in lower-dimensional spaces, and its simplicity as a representative model for human cognition [98]. We believe that this disparity indicates that the application of Bayesian latent dynamics has not been fully explored in mutual adaptation and influence. This may be due to the fact that, compared to Markovian and encoded latent dynamics models, Bayesian models are less personalizable and provide fewer options to tune the model for different tasks and scenarios, which makes Bayesian models less enticing for researchers who are exploring the potential of latent dynamics modeling in mutual adaptation and influence.

Update Frequency: The other interesting observation that can be made by looking at Figure 7 is that the number of articles that used *within* interaction updates is approximately double the number of articles that used *between* interaction updates (Figure 7, Col O-P Row 15-16). One possible reason is that, typically, it is easier to perform within interaction studies as they are more sample-efficient. Also, when modeling the latent update as occurring between interactions, it may be a strong assumption to assert that the human's latent state is stationary within an episode depending on the setting. However, between-interaction updates may afford an opportunity for longer-term influence [37], but this has yet to be thoroughly explored.

Types of Influence: We also observe two types of influence that arise in the reviewed works. First, influence can be *indirect* where the robot attempts to manipulate the latent state as a *means to an end* and only cares about the final outcome of the interaction. This can be seen when the latent state is not directly considered in the reward function but is dynamically modeled during the control step (e.g., you should not care what your air hockey opponent's latent strategy is so long as it allows you to score more goals [22]). Alternatively, influence can be more *direct* where the robot intentionally drives the human to a desired latent state (e.g., modulating trust [20]) and is achieved by adding the latent state as a reward parameter. Deciding which approach to influence is best depends on the interactive goals; is the goal to simply improve robot task performance or elicit a particular quality or belief for the human?

Evaluation Methods: The success of a given mutually adaptive or influential model was found to be evaluated in four ways: task performance, model validity, tractability, and generalizability.

Almost ubiquitously, the reviewed works rely on objective measures of *task performance* (i.e., the ability to interact effectively and achieve task goals) such as reward. Typically, performance was then compared to baseline algorithms, such as Oracle [22, 64] or standard Q-learning [36], to show whether their model led to improved outcomes within a specific task. Task performance metrics commonly include success rate, response time, path tracking error, and perceptual measures like trust and collaboration. Since measuring task performance inherently depends on the nature of the task, there are no standard metrics that would be applicable to all of the reviewed works. However, using statistical significance tests to show a significant improvement in task performance over an existing baseline provides good evidence that mutual adaptation and influence methods can translate to real-world benefits.

As latent dynamics models refer to a hidden variable, it may also be helpful to evaluate *model validity* (i.e., the coherence between the predicted and true latent state). However, it is not always easy or necessary to directly verify that the predicted latent state accurately recovers the ground truth. In mutual adaptation, this is typically not done as the focus is on model performance, not latent state elicitation. Although some works do validate latent state in this way [17], most leverage that fact that adaptability is intrinsically linked to performance [104] and compare to some non-mutually adaptive baseline as in [19]. Similarly, indirect influence (i.e., eliciting a latent state only as a means to an end) usually does not verify latent state accuracy, as it also focuses solely on performance outcomes. However, direct influence (i.e., the goal is to elicit a specific latent state) requires a measurement to validate the model since robot performance is defined by its ability to elicit that latent state. Here, latent states can be measured by induction of empirical observations like policy distributions [22] or through measures of human trust [49, 51], knowledge [21], or perceptions of robot characteristics [41]. Alternatively, some works elect to validate with a simulated human such that the latent state is directly observable [22, 40, 70].

Evaluating the *tractability* of these models has also received some attention. Compared to more standard approaches (e.g., one-way adaptation), introducing latent state dynamics increases computational complexity. While mutual adaptation and influence generally utilize the same evaluation techniques, tractability may be of higher importance for influence due to the repeated application of latent dynamics during the control step. Thus, some works analyze theoretical computational complexity with Big O [6, 20, 55] to ensure tractability. However, most works simply provide empirical evidence of finite time convergence. Evaluating the convergence speed to a stable policy [64] may also be helpful in determining the relative tractability of models given that similar performance is achieved. Similarly, ablation studies [13, 71] are used to evaluate the tradeoff between

computational cost and performance of different latent dynamics models. For example, [29] compared continuous and discrete latent spaces and found that a continuous latent space leads to a noisy reward, which may make learning the optimal policy less tractable. Similarly, [64] discretized a continuous latent space to formulate a tractable problem. While discretization may afford more tractable approaches, higher resolution latent spaces can provide more descriptive power which motivates evaluating latent dynamics models at different task complexities [60].

Finally, evaluating the *generalizability* across individuals and tasks has also received a lot of attention. Simply put, works that study generalization across individuals study how population-level [13, 69], online personalization [17, 63], or sim-to-real [59, 67] models transfer to an unseen partner. More uniquely, a theoretical bound on generalization to an unseen distribution of human partners was given by [67]. Alternatively, some attention is also given to how models transfer to out-of-distribution tasks. This can be studied by evaluating scenarios not observed in the training dataset within a particular task [10, 47] or how well a single model generalizes to a completely new task [26]. Similarly, tasks can be scaled to evaluate how a model performs with different team compositions (e.g., single robot and multi-robot team [44]).

9.3 Open Challenges and Future Work

In this section, we identify some open challenges based on our review, and propose some promising and important future directions for the community.

Standardization and Comparison: The lack of formal standards generally leads to a fragmented evaluation methodology (i.e., tasks and measures) for the field of latent dynamics. A standardized benchmark that fulfills the requirements of many applications is challenging to develop [105] but could better afford meaningful conclusions about the relative effectiveness of different models. This review also did not identify any articles that compared mutual adaptation and influence within the same experimental conditions. In fact, alternatives within any single dimension were not compared as indicated by the upper diagonal in Figure 7. Similarly, with such an agnostic approach to latent dynamics provided by this review, many unexplored areas in both implementation and application are revealed by Figure 7 (e.g., encoded dynamics representations for information exchange and cognitive stimulation in Col K-N Row 19-20). Such comparative evaluations would be helpful to uncover the benefits and challenges of extending mutual adaptation to influence in the real world.

Improving Tractability: As previously mentioned, incorporating latent dynamics introduces additional computational challenges. Primarily, Many of the studies reviewed in this paper formulate the problem as a POMDP, which is known to be intractable in general with respect to computing the optimal policy. To address this challenge, various approximation methods such as the QMDP approxi-

mation [106] have been proposed and demonstrated to be effective in practice [91]. Also, more efficient latent dynamics models can support long-term prediction accuracy and speedup in complex or high-dimensional latent spaces. One promising approach is to apply Neural ODEs for more efficient multi-step predictions by extrapolating future latent states constrained by ODEs instead of repeatedly performing single-step transitions [107]. Additionally, influential policy solutions can be complex when solving the control step with a non-stationary human partner. Thus, applying state-of-the-art methods for solving these non-stationary problems can aid effectiveness of latent dynamics influence and reduce convergence time. For example, sample efficiency of these solutions can be improved with meta-learning [108] for few-shot adaptation of influential strategies or LLM-informed exploration [109, 110] that provide reasonable policy priors.

Improving Generalizability: The majority of existing works formulate task-oriented latent state and dynamics (e.g., navigation goal [42]) or a fixed modality for the dynamics function which limits the range of supported tasks. Thus, models that consider multimodal latent dynamics [111] can better describe variance in dynamics across individuals and tasks, while generative methods like diffusion models [112] may further aid generalization in completely new scenarios, unlike current task generalization, which requires high similarity between training and testing tasks. At the same time, diffusion-based latent dynamics models present a promising direction for modeling multimodal latent state distributions (e.g., multi-modal human belief distributions represented as latent states). They also offer new opportunities for influence, such as steering the sampling dynamics of diffusion models to guide behavior in different directions, as demonstrated in [113]. Leveraging the reasoning capabilities of foundation models may also be an interesting direction for generalization. For example, vision language models (VLMs) [114] may provide a means for generalizing to unseen scenarios by affording zero-shot semantic reasoning about the observations that support latent dynamics. Future work could explore coupling VLMs with latent dynamics models, where the VLM provides high-level semantic embeddings or task descriptions that inform the structure or priors of the latent state space. Such integration could enable agents to reason about latent states or affordances using natural language supervision, paving the way toward more interpretable and generalizable latent representations in human-robot interaction.

Latent Dynamics Alignment: In general, as the name suggests, *latent state* is a hidden variable that cannot be directly observed. In order to effectively track human latent states over time, we need aligned representations for both the latent state and the latent dynamics. *Latent state representation alignment* [16] asks whether the chosen latent state reflects not just functionally descriptive features, but also what the human internally values and models. Such alignment enables models to represent *causal features* of behavior. *Latent dynamics alignment* refers to capturing the *causal relationships* between the latent states, not necessarily if the latent states themselves are descriptive of a

causal feature (e.g., even if a latent state of fatigue is well-represented in a task, a Markovian model might fail to reflect its cumulative, non-Markovian nature, causing poor description of the relationship between one latent state and the next). A misaligned latent dynamics may be able to accurately fit a dataset but fail to capture causal structures, leading to learning spurious correlations and overlooking fundamental processes. A deficit in alignment like this can then degrade value alignment and generalized task learning [16] which highlights the importance of studying latent dynamics alignment. For example, [12] examines human learning in a shared control setting. Here, latent state alignment asks whether the modeled latent state reflects what the human is actually learning about (e.g., the robot's inertia matrix versus its goal position). Likewise, one could do the same with the assumed learning rule (e.g., a Bayesian update or gradient descent) and consider its alignment with the human's actual learning process—this would be referred to as latent dynamics alignment. Without considering both latent state and dynamics alignment, an influential robot can easily mislead the human learner and degrade team performance. While prior work has focused on latent state representation alignment [16, 115, 116], alignment of latent dynamics, to the authors' knowledge, has not been studied and remains an open challenge.

Ethical Considerations: Applying influence in HRI raises several ethical considerations that need to be further studied [99]. For example, interaction data and personalized interaction profiles may contain personally identifiable information and may thus be subject to data privacy protections [117]. Additionally, influence may be applied to drive people to take actions that are not in their best interest, such as utilizing social and emotional mechanisms to make people reliant on a paid service. This is particularly important since humans have been shown to conform to robot suggestions without having a strong basis to trust the robot [118]. Establishing ethical safeguards for transparency, consent, and accountability in robot influence is an urgent and developing area of research. Methods, such as explainable AI, provide intent transparency can help humans understand the motivation behind robot actions and maintain informed consent during long-term HRI.

10 Conclusion

Although robot learning in stationary environments has progressed significantly, interactions with dynamic humans present significant challenges. This review highlights the importance of including latent state dynamics in robots' prediction algorithms for better human-robot interactions in such a non-stationary environment. We proposed a unified framework that extends one-way adaptation, where a robot estimates past human latent states, to mutual adaptation and influence where robots consider the impact of their actions on human behavior. We then used this framework to review compliant mutual adaptation and influence literature and organize the presented approaches into distinct classes of Bayesian, Markovian, and encoded latent

dynamics. Finally, this survey concluded by discussing applications, implementation, emergent properties, and open challenges. In doing so, this paper connects existing work and provides a common framework for classifying, comparing, and integrating latent dynamics models in HRI, thereby supporting a more unified research effort in this field.

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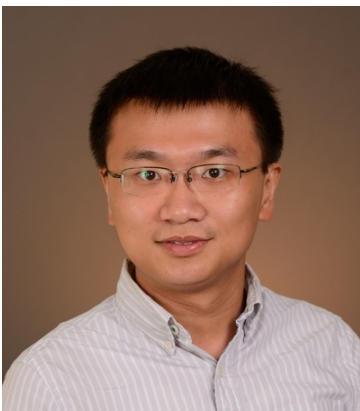
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